Question: 14 (Marks: 1) Attempted Questions: 7 T

In the dining philosopher problem all the philosophers perform two tasks:

Masters

Choices:

- Smiling and laughing
- Thinking and playing
- Thinking and eating

solved by masters

- Eating and playing

Operating Systems (Practical) (CS604P)

Question: 16 (Marks: 1)

A -------- is implemented as a free-space list management.

Choices:

A bit file

A bit map

solved by masters

A block

A stack

An algorithm that replaced the page that has not been used for the longest period of time.

Choices:

- Optimal replacement

solved by masters

- LRU
- FIFO
- LFU

In _____a process must declare in advance the maximum number of instances for each resource type that it may need in the future.

Choices:

- College algorithm
- Lotus algorithm
- Banker's algorithm
- University algorithm

solved by masters

Demand paging technique should ---------- the page fault rate in system.

.....

Choices:

Decrease

solved by masters

.....

Increase

.....

Drop

.....

Fixes

Physical address is computed by appending d (offset number) with f (frame number), size of physical address is calculated by using the formula,

.....

Choices:

- P + d

  solved by masters

  .....
- d × p

  .....
- f + d

  .....
- f × d

Logical data refers to the instruction or data stored in the _____.

Choices:

- Pop table space
- Push table space
- Process address space
- Hash table space

solved by masters

The free space list records all ----------, those not allocated to some file or directory.

.....

Choices:

- (•) Free disk blocks

solved by masters

.....

- ( ) Occupy disk block

.....

- ( ) Accessed disk block

.....

- ( ) Used disk block

Write a c program to calculate the various paging parameters by considering the given values,

Logical Address Space Information: 16
Number of pages: 16
Frames: 32

Calculate the following paging parameters
No. of bits needed for p: ?
No. of bits needed for f: ?

Masters

.....

Answer:

functionalities in the main function of the following code;

- The first loop initializes each semaphore variable i.e. chops[] to 1. Using sem_init() function.
- The second for loop initializes five threads representing the philos_pher(void *) function using pthread_create() function.

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```
#include<stdio.h>
#include<stdlib.h>
#include<pthread.h>
```

Answer:



Write a c program to calculate the various paging parameters by considering the given values,

Logical Address Space Information: 16
Number of pages: 16
Frames: 32

Calculate the following paging parameters
No. of bits needed for p: ?
No. of bits needed for f: ?

Masters

.....

Answer:

written by another program in fifo_one.

- After reading your name and VUID, program1 should display it on the screen by write() system call.

Masters

```
#include <stdio.h>
#include <string.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <sys/errno.h>
#include <stdlib.h>
#include<fcntl.h>
```

See the following C program named program1 that perform inter-process communication through FIFO. Complete the following code and perform the following two tasks.

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- Use the read() system call to read from fifo_one, your name, and VUID that is written by another program in fifo_one.
- After reading your name and VUID, program1 should display it on the screen by write() system call.

```
#include <stdio.h>
#include <string.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <sys/errno.h>
#include <stdlib.h>
#include<fcntl.h>
```

Given the following C program to implement FCFS scheduling algorithm. Given the list of Processes and their burst time, calcula the waiting time for each process and then displays it for each process. Also, calculate and display average waiting time.

Masters

```
#include <stdio.h>
int main(){
int n = 4 ;          // number of processes
int bTime[4] ;    // array for storing burst time for each process
int wTime[4] ;  // array for storing wait time for each process
int tATime[4] ; // array for storing turnaround time for each process
bTime[0] = 10 ;
bTime[1] = 5 ;
bTime[2] = 6 ;
```

```
#include<unistd.h>
sem_t chops[5];
void * philos_pher(void *);
int main()
 {
        int i,n[5];
        pthread_t T[5];
// code for initializing semaphore chops[]


 }
```

Masters